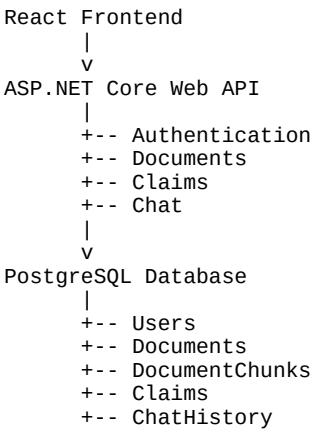
# MediClaim AI v1.0.0

***Professional System Documentation***

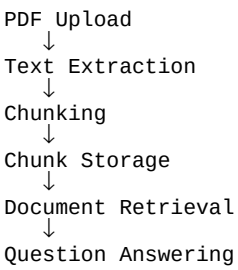
## Executive Summary

MediClaim AI is an AI-powered medical document and claims analysis platform built using React, ASP.NET Core and PostgreSQL. The platform supports secure authentication, PDF document processing, claim risk analysis, dashboard reporting and document-grounded question answering.

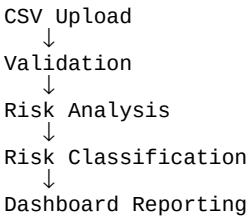
## System Architecture




## Document Processing Flow




## Claims Processing Flow



## Technology Stack

**Frontend**: React, TypeScript, Tailwind CSS, Vite
**Backend**: ASP.NET Core (.NET 10), Entity Framework Core, JWT Authentication
**Database**: PostgreSQL
**Document Processing**: PdfPig, Custom Chunking Engine

## Implemented Features

• JWT Authentication
• User Registration and Login
• PDF Upload and Processing
• Text Extraction and Chunking
• PostgreSQL Storage
• Claims CSV Upload
• Risk Analysis Engine
• Dashboard Analytics
• RAG-Based Chat Assistant
• Chat History Persistence

## Version 2 Roadmap

• Better Citations with Page References
• Prompt Templates
• Embeddings Generation
• pgvector Integration
• Semantic Search
• Hybrid Retrieval
• Evaluation Framework
• Docker Deployment
• Azure Hosting